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No.

ODD END SEMESTER EXAMINATION DECEMBER - 2025

Name of the Course: B.Tech Aerospace Engineering

Name of the Paper: Aircraft Structure-1

Time: 3 Hours

Semester: 5th

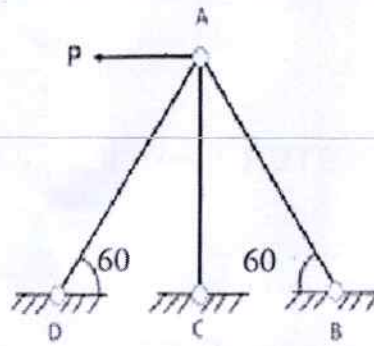
Paper Code: TAS501

Maximum Marks: 100

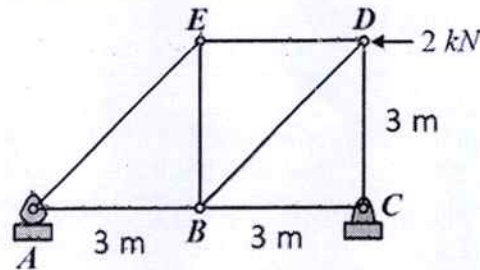
Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1 (20 marks)		CO1
(a)	(i) Develop stress- strain relations for plane stress problems. [5 Marks] (ii) Derive the differential equations of equilibrium and compatibility equations in 2-dimensional Cartesian coordinate system. [5 Marks]	
(b)	At a point in an elastic material there are two mutually perpendicular planes, one of which carries a direct tensile stress at 50 N/mm^2 and a shear stress of 40 N/mm^2 , while the other plane is subjected to a direct compressive stress of 35 N/mm^2 and a complementary shear stress of 40 N/mm^2 . If Young's modlus $E = 210\,000 \text{ N/mm}^2$ and Poisson's ratio $\nu = 0.33$ calculate the direct strain in the x and y directions and the shear strain at the point.	
(c)	At a certain point in a piece of elastic material there are normal stresses of 20 N/mm^2 tensile and 30 N/m^2 compressive on two planes at right angles to one another, together with shearing stresses of 25 N/m^2 on the same planes. If the loading on the material is increased so that the stresses reach K times those given, find the maximum value of K if the maximum direct stress in the material is not to exceed 45 N/m^2 and maximum shear stress not to exceeds 38 N/m^2 . and the maximum shearing stress is not to exceed 40 N/m^2 .	
Q2 (20 marks)		CO2
(a)	Explain the following: stress functions, Mohr's circle, Strain energy and buckling.	
(b)	At a certain point in a strained material the principal stresses are 1200 kg/cm^2 and 500 kg/cm^2 both tensile. Find the normal tangential and resultant stresses across a plane through the point at 48 degrees to the major principal plane, using Mohr's circle of stress.	
(c)	Direct stresses of 120 N/mm^2 tensile and 90 N/mm^2 compressive exist on two perpendicular planes at a certain point in a body. They are also accompanied by shear stress on the planes. The greatest principal stress at a point due to these is 150 N/mm^2 . i) What must be the magnitude of shearing stresses on the two planes? ii) What will be the maximum shearing stress at the point?	
Q3 (20 marks)		CO3
(a)	Prove that the shear stresses are always complementary in nature on two perpendicular plane.	
(b)	Determine the horizontal deformation of joint A of the pin jointed framework as shown below. Assume $A = 1000 \text{ mm}^2$ and $E = 200 \text{ GPa}$ is same for all members and length of the member AC is 500 mm .	



- (c) Explain the term determinacy and indeterminacy. Determine the forces in the member of the pin jointed framework as shown below using nrgy method. Assume A and C are Hinged support and product AE is same for all members.

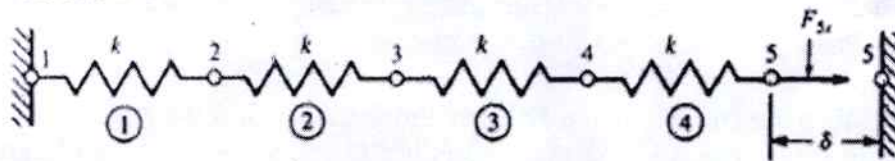


Q4

(20 marks)

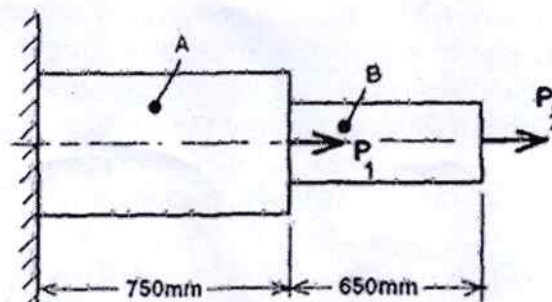
CO4

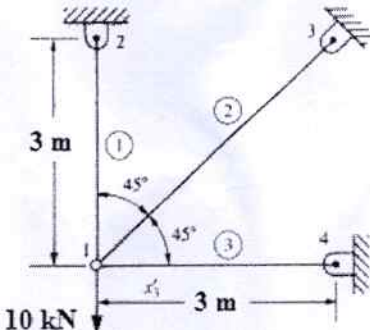
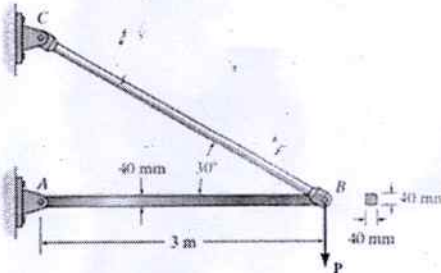
- (a) Consider the following four-spring system; Determine the force at node 5 for the deformation of 20 mm. Also calculate the deformation at each node. Assume spring constant $k = 200 \text{ kN/m}$.



- (b) Determine the nodal displacements, element stresses and support reactions for the stepped bar loaded as shown in Figure P1= 100 kN and P2=75 kN The details of each section of the bar are tabulated below:

Portion	Material	E (GPa)	Area (mm ²)
A	Steel	200	1200
B	Aluminium	70	800



(c)	The plane truss shown below is composed of three bars subjected to a downward force of 10 kN at node 1. Assume the cross-sectional area $A = 1000 \text{ mm}^2$ and E is 200 GPa for all elements. Determine the x and y displacement at node 1. 	
Q5	(20 marks)	
(a)	Define buckling stress? Derive the buckling load for the column with one end is fixed and other is free.	CO5
(b)	The steel bar AB has a square cross section. If it is pin connected at its ends, determine the maximum allowable load P than can be applied to the frame. Use a factor of safety with respect to buckling of 2. 	
(c)	A hollow cylindrical cast iron column is 4-meter-long, both ends are fixed. Design the column to carry an axial load of 25 tonnes. Use Rankine formula and factor of safety 5. The internal diameter may be taken as 0.8 times the external diameter. Take $\sigma_c = 5500 \text{ kg/cm}^2$ and $\alpha = 1/1600$.	

